

Vinh Ly

Palo Alto, CA | vily437@outlook.com | (318) 202-8037 | <https://www.linkedin.com/in/vinh-ly-14b56b50>

Summary

Backend engineer with **18 years** building high-throughput distributed systems across Google, YouTube, and Whatnot. Spent the last several years deep in **ML infrastructure**, recommendation systems, and ad-serving pipelines at YouTube, and before that ran payment processing and Search/Shopping infrastructure at Google. Comfortable presenting complex technical trade-offs to both engineering teams and non-technical stakeholders.

Core Skills

Languages: Java, Python, Go, C++

Frameworks: Spring Boot, Django, TensorFlow, gRPC

Cloud: GCP (App Engine, Kubernetes, BigQuery, Pub/Sub), AWS (EC2, Lambda)

Data & Storage: PostgreSQL, MySQL, Bigtable, Spanner, Cassandra, Redis, Kafka

Observability: Prometheus, Grafana, SLO/SLI monitoring

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Own the ad ranking infrastructure that processes billions of daily requests. The pipeline runs TensorFlow models for real-time bid decisions, and reliability here directly affects revenue.
- Built the ML training platform for YouTube Shorts recommendation models, handling around 40K training jobs per day over roughly 5 PB of daily training data. Cut per-job wall time by about 40% through checkpoint optimization and dataset sharding.
- Worked closely with ML researchers, product managers, and data engineers to design features and validate model behavior before deployment to production traffic.

Staff Software Engineer | Whatnot | Remote | 05/2024 - 02/2025

- Led the migration of the live shopping search and discovery stack from a monolithic Django app into a set of Python and Java microservices. The rewrite was driven by sustained latency spikes during peak sales events, and the new services now handle several thousand live events per second with significantly lower p99 latency.
- Collaborated with the mobile team to design the real-time bidding and notification pipeline, which supported the rollout of a new buyer notification feature.

Staff Software Engineer/Manager | Google | Remote | 03/2015 - 04/2024

- Led a team of 6 engineers maintaining payment processing for Google Search and Shopping, handling millions of transactions per day. Zero unplanned downtime during three consecutive quarters.
- Served as tech lead for the Search infrastructure migration from a custom in-house queueing system to Apache Kafka. The switch simplified ops and reduced tail latency across the pipeline, which processes billions of search result signals daily.
- Reviewed architecture designs and mentored 4 engineers on the team, including two who were promoted to Senior Engineer during that time.
- Managed the team through a reorg that shifted ownership from Search to Shopping, taking on the full payments stack with minimal handover time.

Software Engineer | Google | Hybrid | 02/2011 - 02/2015

- Worked on Search ranking infrastructure, primarily in Java, building features that fed into the core index pipeline. Spent a lot of time on latency profiling and GC tuning to keep p99 request times within SLO.

- Collaborated with search quality engineers to ship ranking changes for three major algorithmic updates.

Software Design Engineer II | Microsoft | Hybrid | 10/2007 - 12/2011

- Worked on the Windows client stack, building features for the shell and file system components in C++. Learned how to write performance-sensitive code and work within a large legacy codebase.

Education

BS Computer Science and Computer Engineering | UC Irvine | 2003 - 2007